



H2S

BHARATIYA ANTARIKSH HACKATHON 2026



Team Name :

UNIONS GIANTS

Team Leader Name :

Abhilasha M Jadhav

Problem Statement :

Rapid Urbanization and climate change have intensified the Urban Heat Island (UHI) effect, causing higher city temperatures, increases energy consumptions, and significant public health risks.

Team Members

Team Leader:

Name: Abhilasha M Jadhav
College: Vidyavardhini's College of
Engineering and Technology (VCET)

Team Member-1:

Name: Vaishnavi Pal
College: Indian Institute of Technology (IIT)
(Banaras Hindu University)

Team Member-2:

Name: Vrushali Bidwai
College: Vidyavardhini's College of
Engineering and Technology (VCET)

Team Member-3:

Name: Yashika Singh
College: Vellore Institute of Technology
(VIT)(Andhra Pradesh)

Why Urban Heat Sentinel ?

a) Existing Problem :

Indian cities are becoming significantly hotter due to the Urban Heat Island effect.

Heatwaves threaten public health, infrastructure, and energy systems.

Current monitoring systems provide limited prediction capabilities.



b) Our Solution & USP :

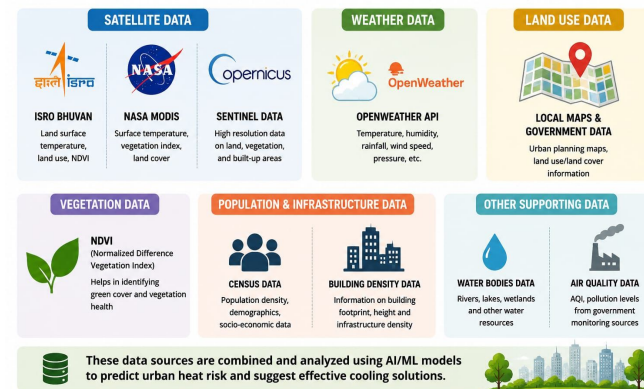
Urban Heat Sentinel integrates:

- . Satellite imagery
 - . AI/ML prediction
 - . IoT weather sensors
 - . GIS mapping
- to detect, predict, and visualize urban heat hotspots in real time.

USP :

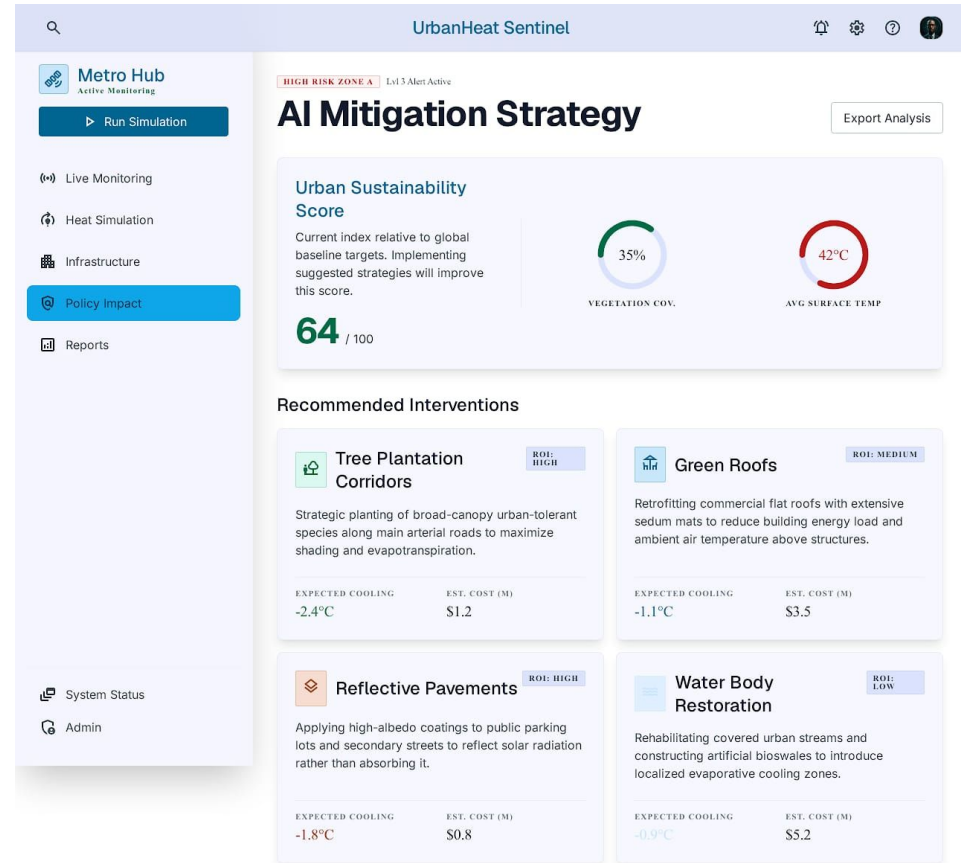
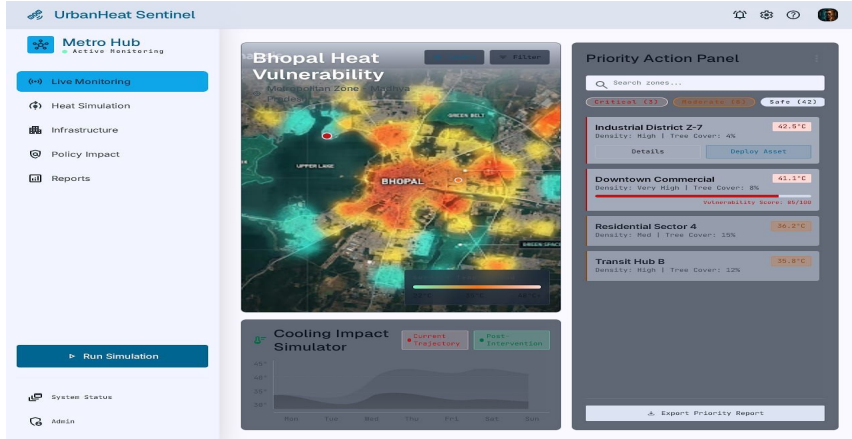
- . AI- Based hotspot Prediction
- . ISRO Satellite Integration
- . Early warning system
- . Interactive heat risk dashboard

DATA SOURCES WE USE

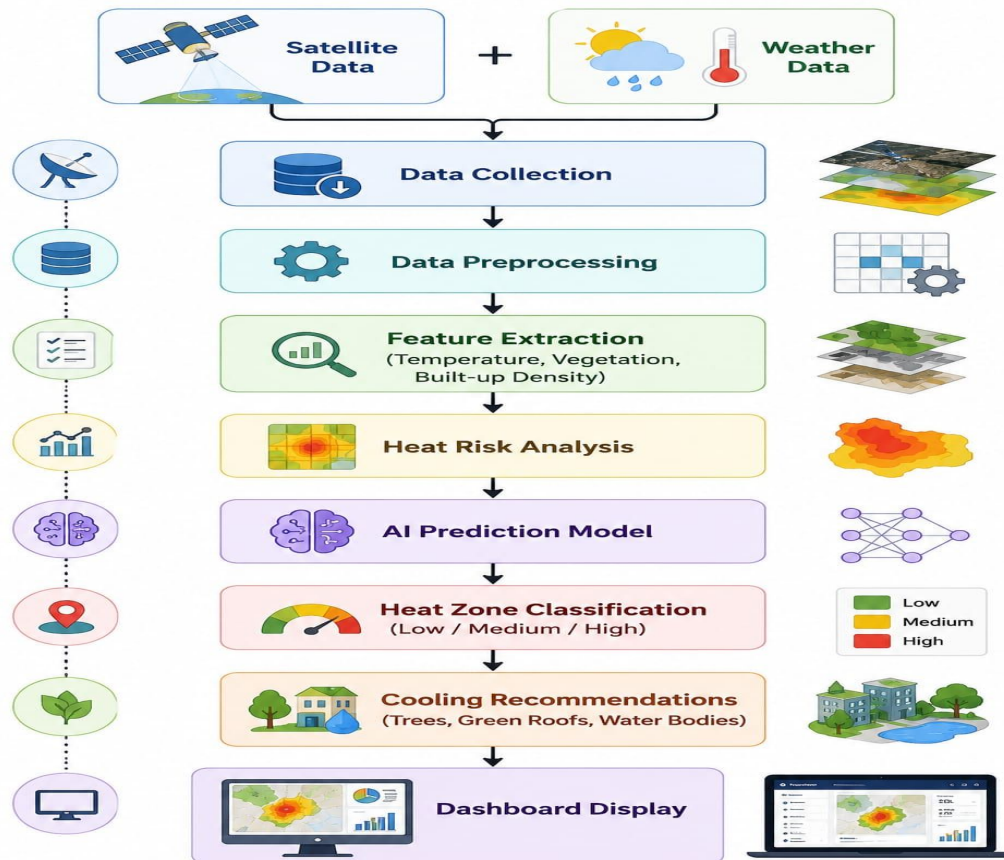


Core Features :

- . Live Urban Heat Index Mapping
- . AI Heat Prediction Engine
- . Citizen Heat Alerts
- . Vulnerability Risk Dashboard
- . Green Zone Recommendation
- . Historical Heat Analytics



FLOW DIAGRAM :



Wireframes/Mock diagrams of the proposed solution :


UrbanHeat AI
v2.4 - AIM Platform

NAVIGATION

Dashboard

Satellite Map

Global Earth Map

Heat Drivers

Scenarios

Interventions

Settings

Overview

Urban Heat Mitigation - AI/ML Intelligence Platform

View Map

New Scenario

Heat Hotspots



7

Active thermal anomalies

Population at Risk



224,200

Residents in hotspot zones

Avg. Temp Reduction



-1.6°C

Across active scenarios

Active Scenarios



2

Running simulations

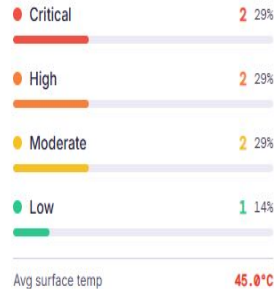
Citywide Temperature Trends

Baseline vs projected vs recorded (°C)



Severity Distribution

Current hotspot breakdown




UrbanHeat AI
v2.4 - AIM Platform

NAVIGATION

Dashboard

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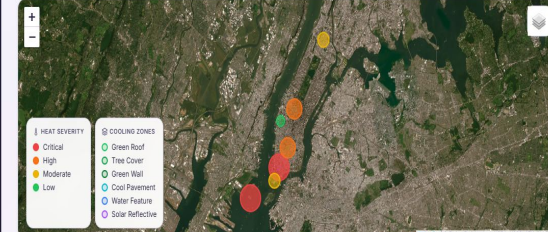
Interventions

Settings

Satellite Thermal Map

Live heat stress hotspots & active cooling intervention zones - ESRI World Imagery

Heat Hotspots Cooling Zones



Heat Hotspots (7 locations)

2 Critical 2 High 2 Moderate

Cooling Zones (5 active)

1 Low 5 Active Zones

-0.9°C Avg Reduction 4.8 Hotspots

Heat Drivers Analysis

Physics-informed ML attribution of urban heating factors

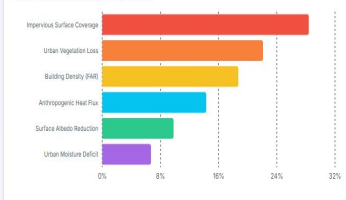
Contribution Breakdown

Relative impact of each heating driver



Factor Contribution (%)

ML-attributed share of urban heat effect



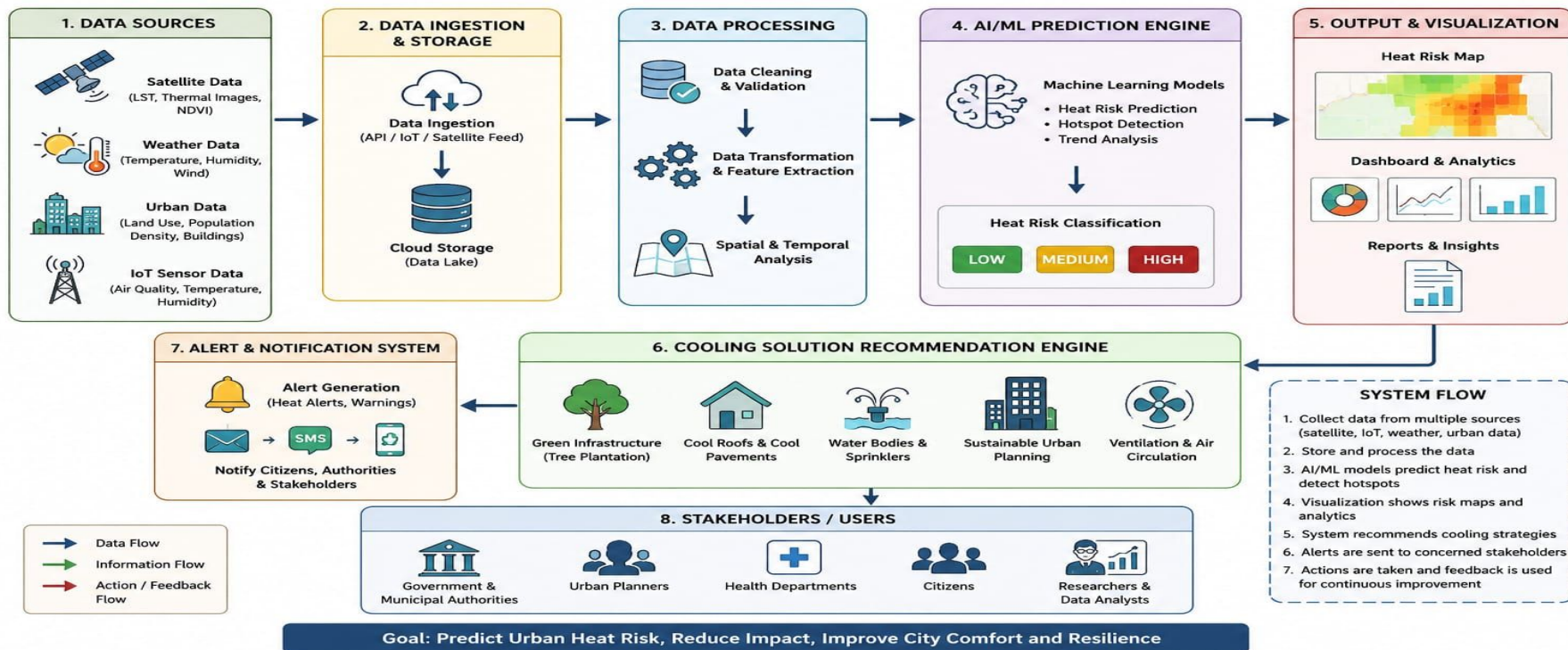
Impervious Surface Coverage **28.4%**
Measured value 72.3 % cover

Urban Vegetation Loss **22.1%**
Measured value 38.6 % canopy loss

Building Density (FAR) **18.7%**
Measured value 4.2 FAR

Architecture diagram of the proposed solution :

ARCHITECTURE DIAGRAM URBAN HEAT RISK PREDICTION AND COOLING SYSTEM



Technologies to be used in the solution:

TECHNOLOGY STACK

Tools & Technologies Used in UrbanHeat Sentinel



HTML

Structure of web pages



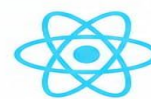
CSS

Styling and responsive design



JavaScript

Interactive and dynamic elements



React

Frontend library for building UI



Python

Core programming language



FastAPI

High-performance API framework



TensorFlow

Machine learning model development



Google Earth Engine

Satellite data processing & analysis



PostgreSQL

Reliable relational database



Leaflet Maps

Interactive maps visualization



ISRO Satellite Data

Real-time earth observation data from ISRO satellites



GitHub

Version control and collaboration platform



Building a smarter, cooler & sustainable future with AI, Satellite & IoT.

Estimated implementation cost :

Development Estimate

<u>Item</u>	<u>Cost</u>
Cloud Infrastructure	: ₹20,000
AI Model Training	: ₹25,000
APIs & Storage	: ₹10,000
Testing & Deployment	: ₹15,000
Total	: ₹70,000

Expected Impact :

Better urban planning
 Reduced heat-related health risks
 Faster emergency response
 Data-driven environmental decisions



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THANK YOU